

Determination of the Physical Size of Galaxies as a Function of Environment and Observational Constraints on the Hubble Constant Using SDSS DR17 Data

K. A. Perez diaz

Master's Degree in Astronomy and Astrophysics

Valencian International University (VIU)

Fecha: 7/08/2026

ABSTRACT

In this work, the linear physical dimensions of galaxies across distinct cosmological distance regimes and environmental densities are determined using spectroscopic and photometric data from the Sloan Digital Sky Survey (SDSS DR17). Samples were isolated from the Virgo cluster ($z = 0.00360$, $N = 97$), Coma cluster ($z = 0.02310$, $N = 328$), and Abell 85 cluster ($z = 0.05506$, $N = 80$), and benchmarked against a distant field galaxy sample ($z > 0.1$, $N = 179$). Under a standard flat Λ CDM cosmological model ($H_0 = 71.0 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_M = 0.27$, $\Omega_\Lambda = 0.73$), angular diameter distances $D_A(z)$ were computed to convert photometric effective radii into intrinsic physical diameters. The median sizes obtained for cluster galaxies range between 3.23 kpc and 5.98 kpc, reflecting the prevalence of compact early-type galaxies modulated by environmental mechanisms within the intracluster medium. In contrast, the field sample exhibits a median of 15.20 kpc, predominantly driven by observational Malmquist bias at high redshift. Furthermore, the functional dependence of physical size on the Hubble constant was investigated over the range $H_0 \in [20, 120] \text{ km s}^{-1} \text{ Mpc}^{-1}$. Benchmarking these scales against reference galaxies in the Local Group (Andromeda, Milky Way, M33, and the Magellanic Clouds) constrains the physically admissible interval to $H_0 \in [50, 85] \text{ km s}^{-1} \text{ Mpc}^{-1}$, in close agreement with constraints derived from the Cosmic Microwave Background (Planck) and local distance-ladder calibrations (SH0ES).

Keywords: Observational cosmology — Hubble constant (H_0) — Galaxy clusters — Angular diameter distance — SDSS DR17.

1. Introducción

Determining the physical dimensions of extragalactic structures represents a fundamental cornerstone of modern observational cosmology. In an expanding universe governed by General Relativity, the size attributed to a galaxy based on its observed angular extent is not a static Euclidean measurement, but an intrinsic quantity inextricably linked to the adopted cosmological framework and the current expansion rate parameterized by the Hubble constant (H_0). Concurrently, galactic morphology and physical extent are modulated by the astrophysical environment: dense cores of massive clusters are governed by hydrodynamic and gravitational interactions, whereas field galaxies evolve in relative isolation. In this context, this paper presents an empirical determination of the physical diameters of galaxies across three clusters spanning different cosmic epochs (Virgo, Coma, Abell 85) and an unbound field sample using spectroscopic and photometric data from SDSS DR17, evaluating environmental impacts and probing sensitivity to cosmological distance scales.

1.1 Theoretical Framework

The modern understanding of large-scale cosmic structure began to emerge in the early decades of the 20th century through the spectroscopic observations of Vesto Slipher (1917) at Lowell Observatory, which revealed systematic redshifts in spiral nebulae corresponding to line-of-sight recession velocities of hundreds of kilometers per second. Simultaneously, Albert Einstein's (1915) General Relativity established the dynamic foundation of spacetime, enabling Alexander Friedmann (1922) and Georges Lemaître (1927) to derive non-static cosmological solutions for a homogeneous and isotropic fluid.

Direct observational confirmation was achieved by Edwin Hubble (1929) via Cepheid variable calibrations in extragalactic nebulae using the 100-inch Hooker telescope, proving their extragalactic nature and establishing a linear relation between apparent recession velocity and distance:

$$v = cz \approx H_0 d$$

Although Hubble's initial determinations substantially overestimated this parameter ($H_0 \approx 500 \text{ km s}^{-1} \text{Mpc}^{-1}$) due to population I and II Cepheid misidentifications (Baade, 1956), the underlying physical principle redefined astronomy: redshift (z) is not a classical kinematic Doppler shift in static space, but the manifestation of metric stretching of null geodesics by cosmic expansion.

In modern cosmology, the standard model is formalized by the Friedmann-Lemaître-Robertson-Walker (FLRW) metric within the flat Λ CDM paradigm. However, the precise value of H_0 remains under intense debate, known as the "Hubble Tension". Local calibrations using Cepheids and Type Ia supernovae yield $H_0 = 73.04 \pm 1.04 \text{ km s}^{-1} \text{Mpc}^{-1}$ (Riess et al., 2022), whereas Planck satellite CMB observations infer $H_0 = 67.4 \pm 0.5 \text{ km s}^{-1} \text{Mpc}^{-1}$ under flat Λ CDM (Planck Collaboration, 2020). This $> 5\sigma$ discrepancy underscores the necessity of assessing how derived physical quantities scale with cosmological parameters.

1.2. Distance Definitions in Relativistic Cosmology

In an expanding metric, Euclidean distance branches into operational definitions tied to specific observables (Hogg, 1999). The cosmic scale factor evolution $a(t)$ is governed by Friedmann's equation:

$$H(z) = H_0 E(z) \tag{1}$$

where the dimensionless expansion function $E(z)$ is given by:

$$E(z) = \sqrt{\Omega_M(1+z)^3 + \Omega_k(1+z)^2 + \Omega_\Lambda} \tag{2}$$

Here, Ω_M , Ω_k , and Ω_Λ denote matter, spatial curvature, and dark energy densities, respectively. Adopting flat spatial geometry ($\Omega_k = 0$), the fiducial model assumes $\Omega_M = 0.27$, $\Omega_\Lambda = 0.73$, and $H_0 = 71.0 \text{ km s}^{-1} \text{Mpc}^{-1}$.

The line-of-sight comoving distance (D_c) is obtained by integrating along the past light cone:

$$D_c(z) = \frac{c}{H_0} \int_0^z \frac{dz'}{E(z')} \quad (3)$$

From this metric baseline, two principal observational quantities emerge:

- **Luminosity Distance (D_L):** Relates intrinsic bolometric luminosity (L) to measured flux (f)

$$f = \frac{L}{4\pi D_L^2} \quad (4)$$

- **Angular Diameter Distance (D_A):** Relates intrinsic linear size (d) to subtended angular angle (θ , in radians):

$$\theta = \frac{d}{D_A} \Leftrightarrow d = \theta \cdot D_A \quad (5)$$

These quantities obey Etherington's reciprocity theorem (Etherington, 1933):

$$D_A(z)(1+z) = D_L(z) = \frac{D_L(z)}{1+z} \quad (6)$$

The angular diameter distance exhibits a characteristic non-monotonic behavior (Weinberg, 1972). At low redshift ($z \ll 1$), D_A increases approximately linearly ($D_A \approx cz/H_0$). At intermediate redshifts, past light cone convergence creates a maximum near $z \approx 1.5$ -- 1.6 . Beyond this turnaround point ($z > 1.6$), $D_A(z)$ decreases, causing objects of fixed physical size to project larger angular diameters on the sky at progressively higher distances.

1.3. Physical Scales of Galaxies and Dependence on H_0

Deriving the physical diameter of a galaxy relies on transforming observed angular diameter θ via D_A . As shown in Equation (3), comoving distance scales inversely with the Hubble constant ($D_c \propto H_0^{-1}$), a property directly inherited by angular diameter distance:

$$D_A(z) \propto \frac{1}{H_0} \quad (7)$$

Consequently, physical size scales as:

$$d = \theta \cdot D_A \propto \frac{1}{H_0} \quad (8)$$

Altering H_0 shifts the metric scale of all distant galaxies. Physical consistency must therefore be evaluated against gravitationally bound Local Group systems (Milky Way, Andromeda, M33, Magellanic Clouds) whose physical dimensions are established via geometric, model-independent methods.

1.4. Objectives of the Study

1. Formulate SQL queries in SDSS DR17 to isolate galaxy samples across four distinct regimes: Virgo ($z \approx 0.0036$), Coma ($z \approx 0.0231$), Abell 85 ($z \approx 0.0551$), and distant field galaxies ($z > 0.10$).
2. Develop a vectorized Python pipeline with `astropy.cosmology` to calculate $D_A(z)$ and convert photometric radii (`deVRad_r`) to physical diameters in kiloparsecs (kpc).
3. Quantify size distributions via publication-grade histograms, medians, and dispersions, characterizing environmental morphology-density relations and Malmquist selection effects.
4. Probe H_0 sensitivity across $H_0 \in [20,120] \text{ km s}^{-1} \text{ Mpc}^{-1}$ and establish empirical constraints using Local Group benchmarks

2. Methodology

The observational and computational workflow comprises three stages: SDSS DR17 catalog extraction, metric size computation via standard relativistic cosmology, and systematic H_0 sensitivity exploration.

2.1. Sample Selection and SDSS DR17 Queries

Samples were isolated across high-density clusters and unbound field environments:

- **Virgo Cluster:** Local virialized system at $z = 0.0036$.
- **Coma Cluster (Abell 1656):** Rich, massive cluster at intermediate distance ($z = 0.0231$).
- **Abell 85 Cluster:** Dense cluster at higher redshift ($z = 0.05506$).
- **Field Galaxies:** Background galaxies selected along a random line of sight, restricted to $z > 0.10$.

For cluster queries, the search radius (θ_{search}) was established assuming a core diameter of 1 Mpc:

$$\theta_{\text{search}} = \frac{1 \text{ Mpc}}{D_A(z_{\text{cluster}})}$$

Spectroscopic catalogs were retrieved from SDSS DR17 (*CasJobs/SkyServer*). Radial kinematic cuts removed interlopers along the line of sight:

$$|c \cdot (z_{\text{galaxy}} - z_{\text{cluster}})| \leq \Delta v$$

using $\Delta v = 2000 \text{ km s}^{-1}$ for Virgo and Coma, and $\Delta v = 3000 \text{ km s}^{-1}$ for Abell 85.

The baseline SQL query joined galaxy photometry and galSpecInfo spectroscopy via dbo.fgetNearByObjEq:

```
SELECT top 500 p.ra, p.dec, p.deVRad_r, q.z
FROM galaxy p, dbo.fgetNearByObjEq(RA_CENTRO, DEC_CENTRO, RADIO_ARCMIN) n, galSpecInfo q
WHERE p.objid = n.objid
AND p.specObjID = q.specObjID
AND p.specObjID <> 0
AND q.z > 0
AND abs(300000.0 * (z_CUMULO - q.z)) < DELTA_V;
```

For the field galaxy sample, the query omitted the velocity constraint and applied a lower cut on redshift ($q.z > 0.10$), ensuring a distant background population decoupled from the local universe.

2.2. Photometric Parameters and Angular Diameter Derivation

The de Vaucouleurs radius in the SDSS r -band (deVRad_r) was used as the tracer of the angular size for each galaxy. For elliptical profiles described by de Vaucouleurs' $r^{1/4}$ law, this parameter corresponds to the effective or half-light radius (r_e), which encloses 50% of the total light emitted by the system.

Since the survey database reports this quantity as a radius measured in arcseconds (arcsec), the total angular diameter of aperture (θ , spanning the full extent of the profile) was converted to radians via:

$$\theta = \left(2 \cdot \text{deVRad_r} \cdot \frac{1^\circ}{3600''} \cdot \frac{\pi}{180^\circ} \right) \text{ rad}$$

2.3. Computational Pipeline and Cosmological Framework

Numerical catalog processing was implemented in Python using the astropy.cosmology package. We adopted a standard flat Λ CDM cosmological model with concordance parameters: $\Omega_M = 0.27$, $\Omega_\Lambda = 0.73$, and $H_0 = 71.0 \text{ km s}^{-1} \text{ Mpc}^{-1}$.

For each individual galaxy with a spectroscopic redshift z , the pipeline executed the following steps in a vectorized routine:

1. **Comoving distance (D_c):** Numerical integration of the expansion history between 0 and z :

$$D_c(z) = \frac{c}{H_0} \int_0^z \frac{dz'}{\sqrt{\Omega_M(1+z')^3 + \Omega_\Lambda}}$$

2. **Angular diameter distance (D_A):** Relativistic metric correction accounting for cosmic expansion:

$$D_A(z) = \frac{D_c(z)}{1+z}$$

3. **Physical linear diameter (d):** Direct dimensional conversion to kiloparsecs (kpc):

$$d = \theta \cdot D_A(z)$$

2.4. Sensitivity Analysis Against the Hubble Constant (H_0)

To evaluate how cosmological scaling influences derived physical sizes, we developed an analysis pipeline where the Hubble constant was systematically varied across the interval from 20 to $120 \text{ km s}^{-1}\text{Mpc}^{-1}$ in steps of $10 \text{ km s}^{-1}\text{Mpc}^{-1}$.

For each value of H_0 , all angular diameter distances D_A and physical diameters d were recomputed across the four samples. The evolution of the resulting medians was benchmarked against the direct physical reference scales of the Local Group: M31 (40 kpc), the Milky Way (30 kpc), M33 (15 kpc), and the Small Magellanic Cloud (5 kpc).

3. Results and Discussion

This section presents the quantitative findings derived from the spectroscopic and photometric processing of the four analyzed samples. First, size distributions are assessed under the fiducial standard cosmological framework; second, we examine the systematic response of galaxy diameters to variations in the Hubble constant (H_0) alongside comparisons with Local Group benchmarks.

3.1. Metric Properties Under the Standard Cosmological Model

The SQL query pipeline isolated statistically significant samples exceeding the minimum requirement of 50 objects per catalog. Table 1 summarizes the fundamental statistical properties retrieved for each galaxy population.

Table 1. Statistical summary of the galaxy samples analyzed under standard Λ CDM cosmology ($H_0 = 71.0 \text{ km s}^{-1}\text{Mpc}^{-1}$, $\Omega_M = 0.27$, $\Omega_\Lambda = 0.73$).

Catalog / Sample	Nominal Redshift (z)	Number of Galaxies (N)	Median Diameter [kpc]	Standard Deviation (σ) [kpc]
Virgo Cluster	0.00360	97	3.23	2.90
Coma Cluster	0.02310	328	5.98	4.67
Abell 85	0.05506	80	5.49	7.17
Field Galaxies	> 0.1000	179	15.20	32.10

3.2. Physical Size Distributions and Environmental Effects

Figures 1 through 4 display the physical diameter histograms obtained for each sample. Each panel indicates the median value alongside a shaded interval representing the dispersion range ($\pm 1\sigma$).

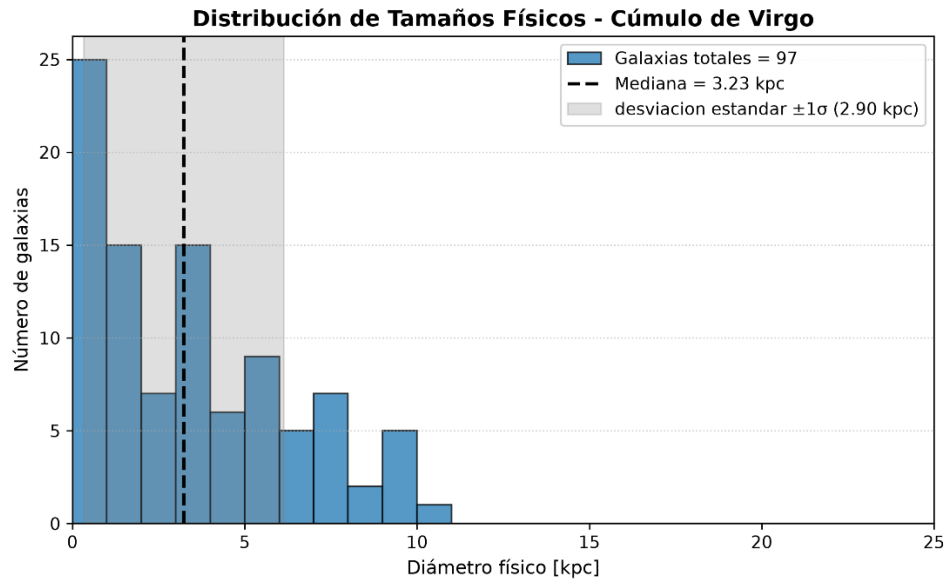


Figure 1. Physical diameter distribution histogram for the Virgo Cluster.

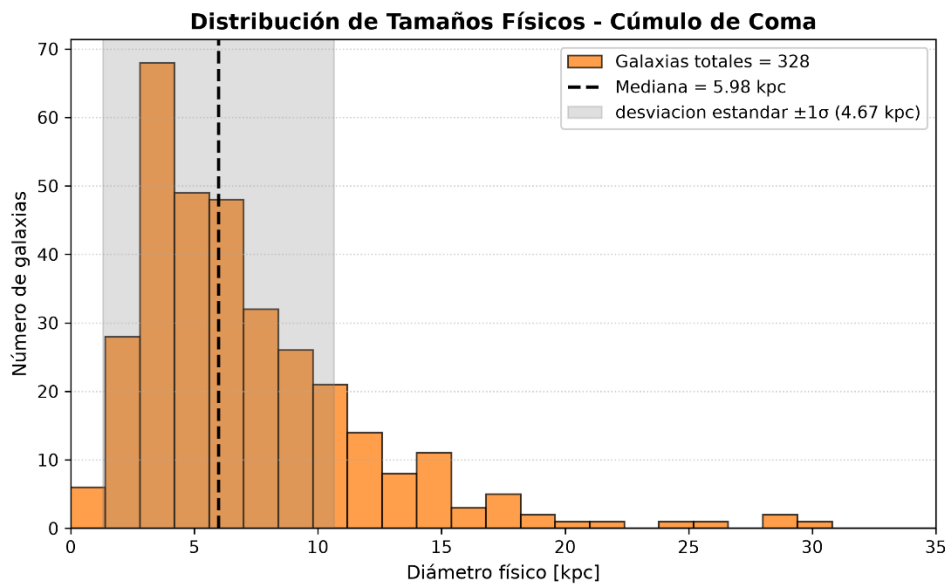


Figure 1. Physical diameter distribution histogram for the Coma Cluster.

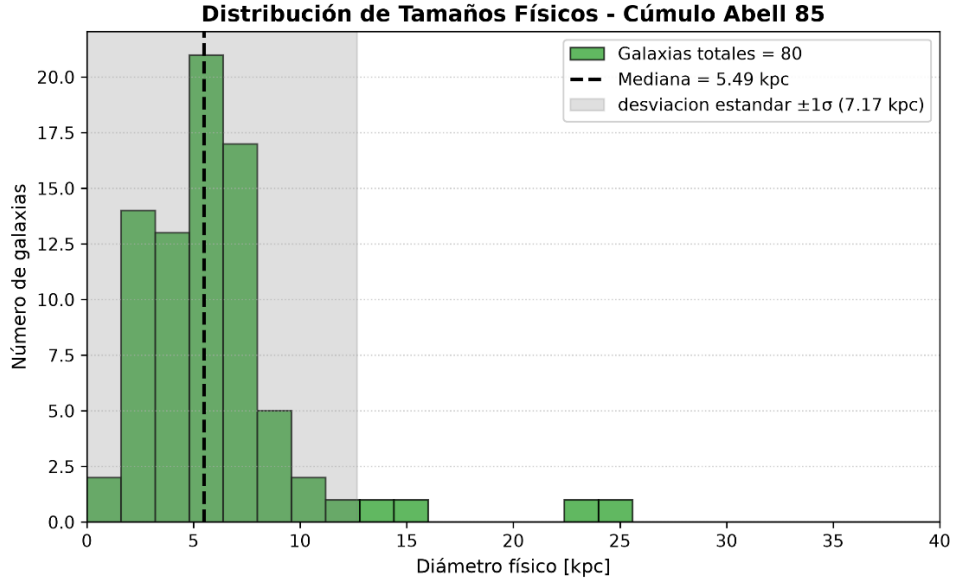


Figure 3. Physical diameter distribution histogram for the Abell 85 Cluster.

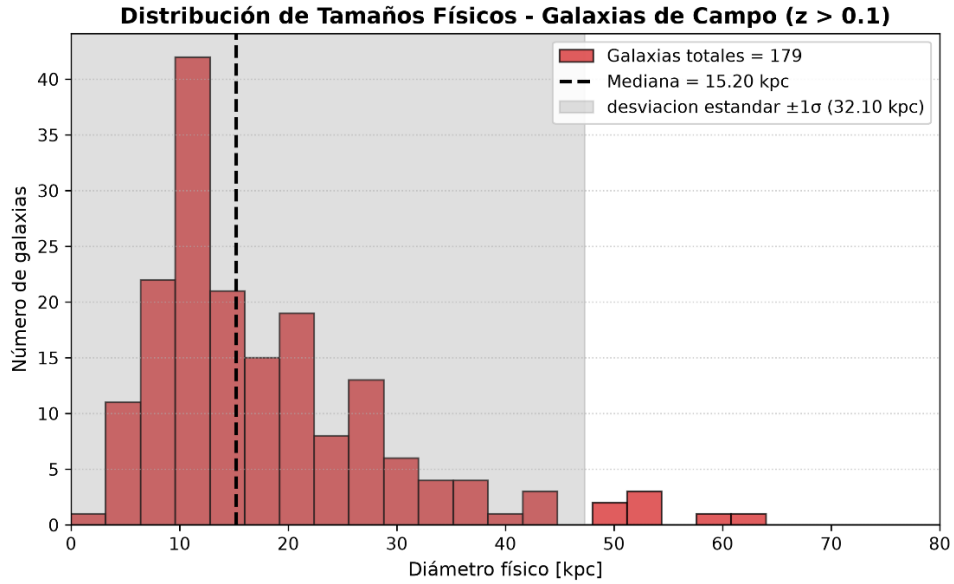


Figure 4. Physical diameter distribution histogram for Field Galaxies.

Based on the analyzed data and morphological inspections of the distributions, three fundamental physical aspects emerge:

- **Morphological contrast between cluster and field:**

Galaxies associated with virialized clusters exhibit markedly compact median physical sizes (3.23--5.98 kpc), substantially lower than the median obtained for the field population (15.20 kpc). This behavior is attributable to morphological segregation effects characteristic of dense environments, where the fraction of early-type elliptical and lenticular galaxies increases systematically over disk spirals (Dressler, 1980). In these regimes, the intracluster medium (ICM) subjects members to hydrodynamic ram pressure stripping capable of removing the gaseous envelopes of disks (Gunn & Gott, 1972), while tidal perturbations from high-velocity encounters (galaxy harassment) truncate outer halos and reshape galactic structure (Moore et al., 1996). The combination of both mechanisms inhibits

extended star formation, favoring the dominance of early-type ellipticals and compact dwarfs characterized by reduced effective radii.

- **Impact of Malmquist selection bias:**

The large difference between the field median (15.20 kpc) and the cluster samples is predominantly driven by an intrinsic observational selection bias. Because the field catalog was restricted to $z > 0.10$, the apparent magnitude limit in the spectroscopic selection of the SDSS preferentially targets the intrinsically brightest sources of the galaxy luminosity function (York et al., 2000). Low surface brightness dwarf galaxies that are readily detectable in local clusters like Virgo fall below the spectroscopic detection threshold at $z > 0.10$.

- **Instrumental sensitivity and dispersion in distant clusters:**

The Virgo Cluster ($z \approx 0.00360$) displays the lowest median size (3.23 kpc) because its proximity allows sampling of a substantial fraction of sub-kiloparsec dwarf galaxies. In contrast, Coma ($z \approx 0.02310$) and Abell 85 ($z \approx 0.05506$) show a stabilization in their medians (~ 5.5 -- 6.0 kpc), consistent with the collapsed core of massive clusters. Abell 85, however, exhibits significantly larger dispersion ($\sigma = 7.17$ kpc), arising from the presence of dominant giant elliptical galaxies in the core (cD-type), visible in the extended tail of the histogram toward diameters exceeding 25 kpc.

3.3. Dependence on the Hubble Constant (H_0) and Observational Constraints

To examine the response of the intrinsic diameter to the expansion rate, the median of each distribution was recomputed by varying the Hubble constant across the range $H_0 \in [20, 120]$ km s⁻¹Mpc⁻¹. The results are detailed in Table 2.

Table 2. Evolution of the median physical diameter as a function of the Hubble constant.

H₀	virgo median (kpc)	coma median (kpc)	abell85 mediana (kpc)	Field median (kpc)
20.0	11.46	21.24	19.48	53.94
30.0	7.64	14.16	12.99	35.96
40.0	5.73	10.62	9.74	26.97
50.0	4.58	8.50	7.79	21.58
60.0	3.82	7.08	6.49	17.98
71.0	3.23	5.98	5.49	15.20
80.0	2.87	5.31	4.87	13.49
90.0	2.55	4.72	4.33	11.99
100.0	2.29	4.25	3.89	10.79
110.0	2.08	3.86	3.55	9.81
120.0	1.91	3.54	3.25	8.99

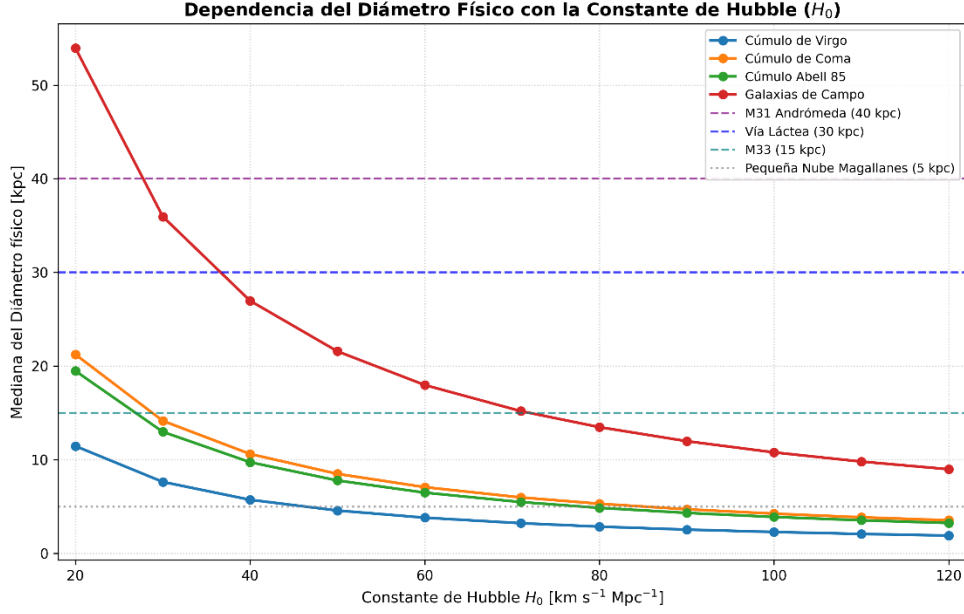


Figure 5. Evolution of the median physical diameter as a function of H_0 in contrast with Local Group reference galaxies.

In accordance with theoretical derivations ($d \propto H_0^{-1}$), Figure 5 displays an asymptotic decrease in physical dimensions as H_0 increases. Confronting these curves with the absolute dimensions of the Local Group allows us to delineate physically viable ranges:

- Infeasibility of low values ($H_0 \leq 40 \text{ km s}^{-1} \text{Mpc}^{-1}$):

For $H_0 \leq 30 \text{ km s}^{-1} \text{Mpc}^{-1}$, the median diameter of field galaxies reaches between 35.96 kpc and 53.94 kpc. This would imply that an average field galaxy in the distant universe possesses dimensions comparable to or exceeding Andromeda (M31, 40 kpc), a physically implausible scenario considering that M31 is an unusually massive giant spiral. Furthermore, cluster galaxies would exhibit excessively high medians ($\sim 13\text{--}21 \text{ kpc}$), erasing the morphological differentiation from mature disk systems.

- Infeasibility of high values ($H_0 \geq 100 \text{ km s}^{-1} \text{Mpc}^{-1}$):

For values of $H_0 \geq 100 \text{ km s}^{-1} \text{Mpc}^{-1}$, the median sizes for the Virgo, Coma, and Abell 85 clusters drop below 4 kpc, with Virgo falling to 1.91--2.29 kpc. This would require classifying the vast majority of the observed cluster population as sub-dwarf systems smaller than the Small Magellanic Cloud (SMC, 5 kpc), contradicting the presence of luminous spectra characteristic of regular ellipticals and spirals.

- Allowed observational interval:

The regime where the medians of dense clusters align with dwarf and intermediate galaxy scales (4--8 kpc) and the field population aligns with intermediate disk galaxies comparable to the Milky Way and M33 (15--30 kpc) is constrained to the interval:

$$H_0 \in [50.0, 85.0] \text{ km s}^{-1} \text{Mpc}^{-1}$$

This interval is consistent with high-precision measurements in modern observational cosmology, such as the Cosmic Microwave Background constraints from the Planck mission ($H_0 = 67.4 \pm 0.5 \text{ km s}^{-1}\text{Mpc}^{-1}$) and the local distance ladder calibration via Type Ia Supernovae from the SH0ES consortium ($H_0 = 73.04 \pm 1.04 \text{ km s}^{-1}\text{Mpc}^{-1}$).

4. Conclusions

Based on the quantitative analysis of the photometric and spectroscopic data extracted from SDSS DR17 and its integration into the standard cosmological framework, the main conclusions are:

- **Analytical Sensitivity of Physical Diameters to the Hubble Constant:**

We empirically confirmed that derived physical sizes scale strictly inversely with the expansion rate ($d \propto H_0^{-1}$). Low values of H_0 systematically overestimate galaxy dimensions to levels inconsistent with the local luminosity function, whereas high values of H_0 compress the observed galaxy population into sub-dwarf regimes incompatible with their luminous spectral classifications.

- **Empirical Constraints on the Parameter H_0 :**

Benchmarking the median sizes against the direct, geometrically calibrated scales of the Local Group (which are independent of the Hubble flow: M31 at 40 kpc, Milky Way at 30 kpc, and SMC at 5 kpc) constrains the physically viable range of the Hubble constant to:

$$H_0 \in [50.0, 85.0] \text{ km s}^{-1}\text{Mpc}^{-1}$$

This interval reliably brackets the range established by state-of-the-art cosmological probes, successfully encompassing both the early-universe Cosmic Microwave Background determination by the Planck collaboration ($67.4 \pm 0.5 \text{ km s}^{-1}\text{Mpc}^{-1}$) and the local distance ladder measurement from the SH0ES consortium ($73.04 \pm 1.04 \text{ km s}^{-1}\text{Mpc}^{-1}$).

- **Environmental Influence on Galactic Morphology:**

Galaxies residing within high-density virialized clusters exhibit markedly compact median physical diameters (3.23 kpc in Virgo, 5.98 kpc in Coma, and 5.49 kpc in Abell 85) relative to isolated field systems. This outcome supports standard models of environmental evolution: hydrodynamic ram pressure stripping and high-velocity tidal interactions (*galaxy harassment*) systematically remove extended disk gas and truncate outer halos, driving the dominance of early-type ellipticals and compact dwarfs with reduced effective radii.

- **Instrumental Impact of Malmquist Selection Bias:**

The significantly higher median size derived for field galaxies (15.20 kpc) compared to local cluster members demonstrates the strong impact of spectroscopic flux limits in survey data. By sampling galaxies at higher redshifts ($z > 0.10$), SDSS spectroscopic completeness is restricted to the most intrinsically luminous and extended systems, effectively filtering out the faint dwarf population that volumetrically dominates nearby clusters.

References

- Baade, W. (1956). The period-luminosity relation of the cepheids. *Publications of the Astronomical Society of the Pacific*, 68(400), 5-16.
- Einstein, A. (1915). Die Feldgleichungen der Gravitation. *Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften*, 844-847.
- Etherington, I. M. H. (1933). On the definition of distance in general relativity. *The Philosophical Magazine*, 15(96), 761-773.
- Friedmann, A. (1922). Über die Krümmung des Raumes. *Zeitschrift für Physik*, 10(1), 377-386.
- Hogg, D. W. (1999). Distance measures in cosmology. *arXiv preprint astro-ph/9905116*.
- Hubble, E. (1929). A relation between distance and radial velocity among extra-galactic nebulae. *Proceedings of the National Academy of Sciences*, 15(3), 168-173.
- Lemaître, G. (1927). Un Univers homogène de masse constante et de rayon croissant rendant compte de la vitesse radiale des nébuleuses extra-galactiques. *Annales de la Société Scientifique de Bruxelles*, A47, 49-59.
- Planck Collaboration. (2020). Planck 2018 results. VI. Cosmological parameters. *Astronomy & Astrophysics*, 641, A6.
- Riess, A. G., et al. (2022). A Comprehensive Measurement of the Local Value of the Hubble Constant with 1 km s⁻¹ Mpc⁻¹ Uncertainty from the Hubble Space Telescope and the SH0ES Team. *The Astrophysical Journal Letters*, 934(1), L7.
- Slipher, V. M. (1917). Radial velocity observations of spiral nebulae. *The Observatory*, 40, 304-306.
- Weinberg, S. (1972). *Gravitation and Cosmology: Principles and Applications of the General Theory of Relativity*. John Wiley & Sons.
- Dressler, A. (1980). Galaxy morphology in rich clusters: Implications for the formation and evolution of galaxies. *The Astrophysical Journal*, 236, 351-365.
- Gunn, J. E., & Gott, J. R. (1972). On the infall of matter into clusters of galaxies and some effects on their evolution. *The Astrophysical Journal*, 176, 1-19.
- Malmquist, K. G. (1922). On some relations in stellar statistics. *Meddelanden fran Lunds Astronomiska Observatorium Serie I*, 100, 1-52.
- Moore, B., Katz, N., Lake, G., Dressler, A., & Oemler, A. (1996). Galaxy harassment and the evolution of clusters of galaxies. *Nature*, 379(6566), 613-616.
- York, D. G., et al. (2000). The Sloan Digital Sky Survey: Technical summary. *The Astronomical Journal*, 120(3), 1579-1587.

Note: Translated from the original Spanish manuscript with the assistance of artificial intelligence.